Computing Machinery and Intelligence

A critical analysis of Alan Turing's landmark 1950 paper

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The Imitation Game

Turing starts with the question "Can machines think?" â€“ and then reformulates it. Defining "machine" and "think" strictly is almost impossible, so he proposes a game.

- **Original Setup:** A judge converses via text with a man (A) and a woman (B), trying to identify who is who.
- **Turing's Twist:** Replace a human with a machine. If the judge makes the same identification errors, the machine has "passed" the test.
- **The Logic:** If you cannot distinguish the machine from a human being in a conversation, it is reasonable and logical to state that it "thinks".

Objections Turing Addresses

Section 6 of the paper is dedicated to anticipating criticisms. Turing directly addresses 5 main objections:

- **Theological:** "Only God can give a soul." → *Turing:* This arbitrarily limits divine omnipotence; God could animate a machine.
- **GÃ¶del / Mathematical:** Formal systems have provable limits. → *Turing:* Human beings make mistakes too, we are not infallible.
- **Consciousness (Jefferson):** Machines cannot truly "feel". → *Turing:* This leads to solipsism: we cannot even verify with certainty if other humans feel emotions.
- **Lady Lovelace:** "Machines only do what they are told to do." → *Turing:* Machines surprise him continuously; furthermore, it is difficult to distinguish where education ends and originality begins (concept of Learning Machines).
- **Informal Behavior:** Not all human behavior can be codified. → *Turing:* Partially true, but this does not prevent a machine from passing the test in practice.

My Criticisms of the Paper

Analyzing the test through a modern lens, several critical vulnerabilities emerge:

- **Imitation vs. Understanding:** The test replaces "thinking" with "appearing human". Modern models can respond convincingly by drawing from immense databases, without possessing any real understanding (Stochastic Parrots).
- **Human-Centered Bias:** It implicitly assumes that intelligence must necessarily resemble human intelligence. A radically different intelligence (e.g., alien or spatial) would fail the test despite potentially being immensely superior.
- **Hasty GÃ¶del Response:** Turing's response is too hasty. GÃ¶del's theorem concerns the structural and mathematical limits of logic, not the simple "human tendency to err".
- **Vague Pass Criteria:** Turing speaks of 70% indistinguishability, but does not specify the number of judges, the duration of the conversation, or the variety of topics, making the test impossible to replicate with scientific rigor.

When Can We Say the Test Has Been Passed?

Assuming the test is meaningful, we identify 6 rigorous criteria to pass it today:

- **Statistical Threshold:** Judges correctly identify the machine less than 50% of the time over a large statistical sample, not in a single experiment.
- **Robustness to Tricks:** The machine must resist trick questions: logical paradoxes, physical sensations, and recent events outside its training.
- **Qualified Blind Judges:** Judges must know AI to avoid naivety, but the test must remain strictly "blind" (they must not know in advance who they are interrogating).
- **Topical Variety:** Questions must range across logic, emotions, humor, ethics, and culture â€" not be limited to arithmetic or encyclopedic facts.
- **Longitudinal Coherence:** The machine must not contradict itself, must remember previous statements, and maintain a stable personality over time.
- **Unseen Questions:** Questions must never have appeared in the training data. Otherwise, the test measures memorization and not intelligence.

THANK YOU!